



Sachin Kumar

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Career Objective

I'm hardworking, technically astute, and can work independently or in a team. I possess good communication skills, both verbal and written and always strive to meet deadlines. I have professional dedication and a positive approach with a learning attitude toward newer technologies and accepting challenges.

Education

Center for Converging Technologies, Jaipur

- *Dual Degree B. Tech M. Tech in Nanotechnology* (July 2016 - 2021)

This course is about converging technologies in which I've studied the convergence of two or more technologies. I've studied **Biotechnology and Bioinformatics** as a major subject and **Neuroscience** as a minor subject. I graduated in nanotechnology and scored a **5.09 CGPA out of 6**.

Bal Bharti Sen. Sec. School, Sadulshahar, Sri Ganganagar

- *Senior Secondary* (July 2015 - May 2016)

Here, I studied Physics, Chemistry, and Biology for the senior secondary examination, and I got **70%** in the state board examination.

- *Secondary* (2013-2014)

I got **57.17%** in the state board examination.

Skills

- **Molecular techniques:**

- DNA Extraction (blood, cells, Water, Soil), RNA extraction (Blood)
- PBMC isolation (blood and cells), cDNA, Protein isolation (cells), Mitochondria isolation
- handling of a blood sample, handling of the covid-19 sample, animal handling, Processing of Covid 19 samples

- **Cell culture:**

- Cells handling (A549, N2A, sf9)
- Media preparation, cell culture, trypsinization, subculture
- MTT assay, Mitochondrial cytokines assay

- **Microbiology Experiment:**
 - Media Preparation, plating, bacterial isolation, Antibiotic Substitutability test, Bacterial handling, Colony count, Bacterial Growth (E. coli, S.Aurios), bacterial Conjugation
- **Animal Handling and Experimentation**
 - Tail amputation
 - Wound healing
- **Technical skills**
 - PCR, RT-PCR, qPCR, Digital PCR, ELISA Assay, Gel Electrophoresis (Agarose, SDS-PAGE), Western blotting, U.V. Spectrophotometer, Nanodrop, Laminar Airflow, Centrifuge, Autoclave, fluorescence microscope
- **Computer Knowledge**
 - C, C++, Java, MsOffice. Analysis software, Bioinformatics (BLAST, FASTA, CLUSTALW, MEGA, ORF FINDER, SMILES, VISTA)

Experience

➤ INMAS DRDO, Delhi — *Research Intern*

July 2021 - May 2022

In this period, I've done my M. tech dissertation at the Dept of MARB, INMAS DRDO, Delhi. Here, I have been working on Gene Expression Radiation biodosimetry. My thesis title was "Standardization of RNA extraction from ex-vivo irradiated human blood at different doses and time points" under Dr. Shuchi Bhagi, Scientist- 'C' at the Molecular and Radiation bio dosimetry department."

➤ Datta Meghe Institute of Higher Education and Research, Wardha, Maharashtra --- *Junior Research Fellow*

September 2022 - Feb 2023

At DMIHER, I worked as JRF on the ICMR Project "Bioprinting of skeletal tissue towards customized phalanx reconstruction by tissue engineering approach" under the supervision of Dr Ashutosh Bagde. The project collaborated with Prof. Shantanu Dhara, IIT Kharagpur.

➤ I.I.T. Hyderabad, Telangana— *Junior Research Fellow*

March 2023 – till now

At IIT Hyderabad, I am a JRF on AMR Flows: "Antimicrobial Resistance from manufacturing flows to people: Joined up to experiments and Mathematics modeling and risk analysis." The Project entitled ". I am working as a molecular biologist.

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Projects Seminar

and

- A 2-week Hands-on Training Workshop, Monitoring and characterization @ Metal Additive Manufacturing (05-12-2022 - 18-12-2022)
- AR/VR by AMTZ workshop November 2022
- Attended webinar on “computer-aided drug design,” Jan 2021
- Attended workshop on “Basics of networking including creating a network,” organized by CCT, UOR, Jaipur, May 2019
- Attended lecture series on “Nanotechnology,” October 2018, organized by the Dept. of Chemistry, UOR, Jaipur.

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Languages

- English
- Hindi
- Punjabi

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Publication

1. The Burden of Lifestyle Diseases and Their Impact on Health Service in India” - A Narrative Review DOI: 10.4103/jfmpc.jfmpc_693_23
2. Under Review, a manuscript ID (D-3412022) entitled “Anti-inflammatory herbal medicines for the management of Obsessive-Compulsive Disorder” is submitted to the Biologics and Drug Discovery Journal of Petrichor Biologics Foundation,
3. Under Review a manuscript: Biologics and Drug Discovery Journal Manuscript Submission D-3312022
4. Under review: A Review of Radiation Bio dosimetry Techniques used for Biomarker Profiling

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References

1. Dr Shuchi Bhagi (Scientist ‘C’)
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2. Dr. Amit Alok (Scientist ‘E’)
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3. Prof. P.J. John (‘Director’)
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